# Drashti Ankurkumar Bhavsar

+1 857-397-6380 drashtiabhavsar9@gmail.com linkedin.com/in/drashtibhavsar9 github.com/Drashti0913

Available for a Full-Time Spring/Summer 2027 Internship — Open to Relocation

## EDUCATION

**Northeastern University, Boston, MA** **August 2025 - Expected December 2027**

*MS in Computer Science*

**Pandit Deendayal Energy University, India** **May 2024**

*Bachelor of Technology in Computer Engineering* ***3.78/4.0***

***Coursework:****Data Structures & Algorithms, Object-Oriented Programming, Distributed Systems, Machine Learning*

## TECHNICAL SKILLS

**Languages**: Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS
**Frameworks**: React, Node.js, FastAPI, Django, Flask, Express.js, MATLAB
**Systems**: REST APIs, GraphQL, WebSockets, Microservices, Apache Kafka, Redis, Git, Postman, Jira, Agile/Scrum
**Databases**: PostgreSQL, MongoDB, MySQL, Elasticsearch, Qdrant
**Cloud & DevOps**: AWS(S3, Lambda), GCP, Azure, Linux/Unix, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions
**AI/ML:** PyTorch, TensorFlow, Scikit-Learn, NLP, Computer Vision, NLTK, Codex, Claude Code
**Robotics & AI:** CARLA, PyTorch, TensorFlow, OpenCV, Deep Reinforcement Learning, Computer Vision, Object Detection, Autonomous Navigation

## EXPERIENCE

**LOTI AI Inc.** **August 2024 - July 2025**

***Machine Learning Engineer Intern*** | PyTorch, OpenCV, BGE, Qwen *Seattle, WA*

- Engineered a production Python inference platform integrating multiple APIs with error handling, rate limiting, and monitoring, sustaining 99% uptime across 10K+ daily requests.
- Architected scalable microservices using Docker and Kubernetes for real-time model inference, reducing deployment time from 2 hours to 15 minutes with automated CI/CD pipelines.
- Collaborated cross-functionally with 5-person engineering team using Agile methodology, conducting code reviews and maintaining 95%+ test coverage with comprehensive unit and integration tests.

**IIT Roorkee – Google ExploreCSR** **January 2024 - June 2024**

***Junior Research Fellow*** | Python, PyTorch, OpenCV, CARLA, Deep Reinforcement Learning *India*

- Developed hierarchical deep reinforcement learning models in CARLA for autonomous-vehicle navigation using depth-camera inputs and obstacle-detection modules.
- Engineered state features and reward functions for overtaking, lane changing and collision avoidance, improving navigation decision accuracy by 25% in dynamic environments.
- Evaluated the agent across simulated urban scenarios, achieving a 94% task-success rate with inference latency ¡50ms.

**Indian Space Research Organisation - ISRO** **June 2023 - August 2023**

***Data Science Research Intern*** | Python, Random Forest, LSTM, Forecasting, Classification *Ahmedabad, India*

- Contributed to "Navigation Signal Anomaly Detection" research, performing statistical analysis and time-series forecasting on satellite data using Plotly and Pandas.
- Trained and optimized Random Forest and LSTM models using Scikit-Learn and TensorFlow achieving 98% and 97% accuracy respectively through hyperparameter optimization and data preprocessing.
- Deployed real-time L5-signal classification model using FastAPI with 23% improved positioning accuracy; Analyzed internal network accessibility through ngrok tunneling for production monitoring.

## PROJECTS GitHub

**Real-Time Collaborative Code Editor** | *React.js, Node.js, WebSockets, Redis, PostgreSQL*

- Created real-time editor with WebSockets supporting 50+ concurrent users at 100ms latency implementing operational transformation for 99.9% data consistency.
- Launched backend with Redis and PostgreSQL on AWS EC2 reducing database load by 60%.

**Intelligent Traffic-Aware Delivery Optimization** | *Python, A-star, Dijkstra, Q-Learning, DQN*

- Built a traffic-aware routing system comparing A-star, Dijkstra, Greedy Best-First Search and TSP heuristics across dynamic delivery scenarios.
- Developed Q-learning and DQN agents to adapt routing decisions based on traffic conditions and evaluated total travel time, customer wait time, completed deliveries and runtime.

**YouTube Thumbnail Analyzer Bot** | *Python, Telegram API, YouTube API, asyncio, PIL*

- Created Telegram bot integrating 3 external APIs with OAuth authentication serving 100+ users at 99% uptime.
- Streamlined parallel processing with asyncio handling 20 thumbnails in 60s with automated testing and multi-language.

## PUBLICATIONS Google Scholar

- Health Misinformation Detection in Hidradenitis Suppurativa Communities Using Multi-Model LLM Ensembles
- Milk Quality Prediction Using Machine Learning. ***(EBSCO)***
- AI-enabled dental caries detection using transfer learning and gradient-based class activation mapping ***(Springer)***